

Evaluating ASR Confidence Scores for Automated Error Detection in User-Assisted Correction Interfaces

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Abstract

Despite advances in Automatic Speech Recognition (ASR), transcription errors persist and require manual correction. Confidence scores, which indicate the certainty of ASR results, could assist users in identifying and correcting errors. This study evaluates the reliability of confidence scores for error detection through a comprehensive analysis of end-to-end ASR models and a user study with 36 participants. The results show that while confidence scores correlate with transcription accuracy, their error detection performance is limited. Classifiers frequently miss errors or generate many false positives, undermining their practical utility. Confidence-based error detection neither improved correction efficiency nor was perceived as helpful by participants. These findings highlight the limitations of confidence scores and the need for more sophisticated approaches to improve user interaction and explainability of ASR results.

CCS Concepts

• **Human-centered computing** → **Natural language interfaces**; *Empirical studies in HCI*; *Empirical studies in interaction design*; *Empirical studies in visualization*; *Empirical studies in accessibility*;
• **Computing methodologies** → **Speech recognition**.

Keywords

automatic speech recognition; confidence measures; error detection; keyword highlighting

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1 Introduction

Confidence scores are an additional output of an artificial intelligence (AI) system and represent the degree of certainty a model has in its prediction. In Automatic Speech Recognition (ASR), these scores reflect the probability with which a spoken word is recognised correctly. Despite recent improvements in the accuracy of

ASR [1, 46], transcripts often require manual editing to be intelligible [6, 40, 54]. Automatically highlighting transcription errors could help users to correct them more quickly and accurately. One approach is to use confidence scores to highlight low-confidence words to indicate potential recognition errors [3, 38, 39]. Such automated error detection could help users to correct errors more quickly and accurately.

ASR models can provide word-level confidence scores, which are usually represented as numerical values between 0 and 1, with higher scores indicating greater confidence in the correctness of the result. To automatically classify words as correct or incorrect based on the confidence score, a threshold must be set that defines the minimum confidence level at which a word should be considered correct (e.g. all words above a score of 0.5). As this classification does not necessarily align with the actual correctness of a word, it can lead to false positives and missed detections. Ideally, the threshold is well balanced, assisting users to find major errors without distracting them with non- or minor errors.

Confidence scores are mainly used for downstream machine learning (ML) tasks, but have also been proposed to assist users in correcting transcriptions [38, 39] and d/Deaf and hard of hearing (DHH) individuals in reading automated captions [3, 26]. However, research on their reliability and effectiveness for these tasks remains limited. Evidence from early studies suggests that they are either unreliable [16, 57, 63] or have a limited positive effect on user performance [5]. As the architecture of ASR has changed fundamentally in recent years, findings from earlier evaluations may not be directly applicable to today's systems. Transformers and recurrent neural networks (RNNs) using attention mechanisms have largely replaced hidden Markov model (HMM)-based models in modern ASR [8, 11, 18, 30, 43], leveraging large-scale deep learning techniques and end-to-end (E2E) training [1, 47, 69]. These E2E models show increased robustness and accuracy [46] but use differing confidence estimation techniques [42] and exhibit overconfidence [19]. Although the use of confidence scores in E2E models for ML is an active area of research [13, 32, 37, 58, 66], their application in user interfaces may differ and has not yet been evaluated.

This study explores whether confidence scores can reliably detect transcription errors and assist users in correcting transcripts. We comprehensively analyse current ASR models employing E2E architectures. We complement the quantitative evaluation with a user study applying the confidence-based error classification for manual correction. Our results show:

- Classifiers using confidence scores achieve only moderate performance and cannot reliably detect transcription errors.
- Confidence-based error detection neither improved correction efficiency nor proved useful to participants.

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2 Background

To contextualise previous Human–computer interaction (HCI) research on confidence scores, it is important to understand the development of ASR models and corresponding confidence measures. This section first looks at the technical fundamentals of confidence scores and then at their applied use in HCI studies.

2.1 Confidence Measures in Machine Learning

Confidence measures are used for various ASR downstream tasks, particularly ML methods like unsupervised [69], semi-supervised [7, 24, 42], and active learning [23, 35, 68]. They are also used for speaker adaptation [12, 60], keyword spotting [2, 29, 53], and dialogue systems [20, 59]. In HMM-based ASR, confidence is often predicted using binary classifiers on decoding features such as word posterior probabilities [15], word trellis stability [48], normalised likelihoods, and language model score [65]. These features are classified using simple linear mappings, such as decision trees [15] and Gaussian mixture classifiers [10], or sequence models such as conditional random fields [51], neural networks [49, 64], and recurrent neural networks [27, 28, 36].

ASR systems have evolved from HMM and hybrid designs to E2E models, which rely on an encoder-decoder architecture [8, 14, 30]. A common confidence measure for these models, the softmax probability [42], is often unreliable due to the overconfident behaviour of E2E models [19, 22, 37]. To improve the reliability of confidence scores, a model can be calibrated using temperature scaling [19]. However, mitigating high confidence predictions for Out-of-Domain (OOD) data remains challenging and may be an inherent limitation of neural network architectures [21]. While approaches such as dropout, ensembles, or Bayesian methods can improve uncertainty estimation for some tasks, the accuracy also degrades under dataset shifts [33, 41]. Alternatively, a separate neural network can estimate confidence [37, 66], using features from the encoder, decoder, and attention blocks [32, 56]. It has also been explored to improve confidence scores around word deletions [45, 52].

E2E models have become the industry standard due to their improved accuracy, but they also exhibit overconfidence. Several techniques can improve their confidence estimation, but none has proven superior. Research suggests that confidence measures become less accurate as ASR accuracy decreases. Consequently, transcripts that contain more errors and require more corrections, tend to have less reliable confidence scores.

2.2 Confidence Scores in HCI

Despite improvements in ASR accuracy [1, 46], transcripts often require manual correction to ensure accessibility [6, 40, 54]. Automatically highlighting errors could speed up editing and improve accuracy. Confidence scores can indicate misrecognised words, but as continuous data, they require a threshold to classify words as correct or incorrect. Choosing an appropriate threshold is a trade-off between precision and recall, aiming to minimise false positives and negatives while maximising true positives.

Early ASR systems have already shown a strong correlation between transcription accuracy and confidence scores [17], making confidence-based error detection a long-standing idea [9]. However,

early empirical evaluations suggest that confidence scores are limited in detecting errors and highlight the challenges of determining an effective threshold [16, 57]. While highlighting low-confidence words could slightly improve correction accuracy, editing speed remained unchanged [5]. Error markup also influenced which words users corrected, drawing attention to or away from errors [63].

Alternatively, confidence scores can highlight correctly transcribed words, helping readers focus on keywords while ignoring errors [3]. While users expressed a strong interest in such a feature, they found it neither useful nor disruptive [55], with some preferring no markup after use [4]. Highlighting "correct" words improved readers' confidence and comprehension [44], though other studies found no significant benefits from colour-coded text [62]. Underlining and font weight were preferred as markup styles [25], and highlighting 5-15% of words as important was favoured [26].

Confidence scores have been proposed for correction interfaces [38, 39] and as accessibility markup [3, 26], but evidence supporting these ideas remains limited. Previous studies have highlighted the limitations of then-current HMM-based confidence score implementations [57], called for further research [4, 55, 62], and hypothesised that improved confidence accuracy could enable practical applications [5]. This study contributes to the scarce literature that examines E2E models and their modified confidence measures from an HCI perspective. We provide a comprehensive evaluation of state-of-the-art (SOTA) ASR systems and the results of an applied user study.

3 Methodology

We first transcribed a public dataset with current ASR models that provide word-level confidence scores to evaluate their accuracy for error classification. We then conducted a user study to evaluate the error detection method in practice, its impact on correction efficiency, and its more subjective usefulness. Transcription results, analysis code, and editing prototype are available open-source.¹

3.1 Dataset

We selected a dataset containing 90 English and 30 German recordings, each three minutes long, with an average of 466 words per sample [31]. The English split was used for a large-scale evaluation and the German split for a user study. We consider the dataset as in-domain data, with realistic audio conditions, resulting in average transcription accuracy and confidence scores by ASR systems.

The dataset was transcribed using nine different ASR models: Amazon, AssemblyAI, Deepgram, Google, IBM, RevAI, Speechmatics, SpeechText.AI, and Whisper. To the best of our knowledge, all providers have adopted E2E models in recent years. All settings were unified to offer the highest accuracy, e.g. selecting large or enhanced models.

3.2 Analysis

We used the Levenshtein distance algorithm, which is the basis of the Word Error Rate (WER), to align ASR transcripts to a reference solution at the word level. The alignment results are word pairs with an operation (ok, substitution, insertion and deletion), a reference word (except for insertions) and a hypothesis word with

¹<https://doi.org/10.5281/zenodo.14923041>

confidence values (except for deletions) [34]. For binary classification, "ok" denotes correctness and other operations indicate errors. The 810 ASR transcripts yielded 394,348 word pairs, including 7,489 deletions (which have no confidence score and were excluded from the evaluation). We applied JiWER² text transformations to address non-semantic differences, converting text to lowercase, removing punctuation, and expanding contractions. Although Levenshtein-based classification can mislabel correct words due to nearby errors, we expect these errors to have a very limited effect, given the size of the dataset.

3.3 User Study

3.3.1 Design Rationale. The user study evaluates the effectiveness of confidence-based error markup for transcript correction while minimising confounding variables. A single interface, incorporating findings from prior research, was used. It featured three settings with varying levels of error highlighting: no highlighting, balanced highlighting (balancing correct detections and false positives), and high-threshold highlighting (increasing detections but with more false positives). We used a repeated measures design, where each participant worked with all three interface variants in a randomised order using a Latin square design. Participants corrected transcripts in their native language, German, to avoid biases from second-language use. The task involved real-time editing without audio replay, requiring participants to correct the transcript in one go. This approach was chosen to control for individual effort, preventing it from dominating the results and obscuring the effect of confidence-based error highlighting.

3.3.2 Participants. 36 people, including undergraduates, postgraduates, and researchers from different universities, participated in the study. All were native German speakers, gave their informed consent, and confirmed they had no hearing, visual, reading, or writing impairments.

3.3.3 Transcripts. The transcripts were generated using Whisper, showing high transcription accuracy and average performance for error classification in our evaluation (see Section 4). Whisper is widely used in scientific research and ensures more comparable and reproducible results. We selected three transcripts from the German dataset split (IDs 70, 82, and 89) with an average accuracy (5.5–6.3% WER) and a balanced distribution of substitution, insertion, and deletion errors. Balanced thresholds (0.70–0.88) highlighted about 4% of the words, while a high threshold of 0.99 highlighted 13–21% of the words (cf. [4, 26]).

3.3.4 Interface. The user interface (see Figure 1) featured three markup types: inverted text to indicate the currently spoken words, a wavy red underline for potential transcription errors (similar to spellcheckers), and bold text for user-modified words. Markup styles were informed by prior research [4, 26], accessibility considerations (e.g., high contrast, colour independence), and familiarity with text editing software while avoiding distractions like text jitter. The text editor was built using Tiptap³ with a custom schema to manage confidence scores, timestamps, and correctness for each word. Edits extended this metadata to track modifications, creations,

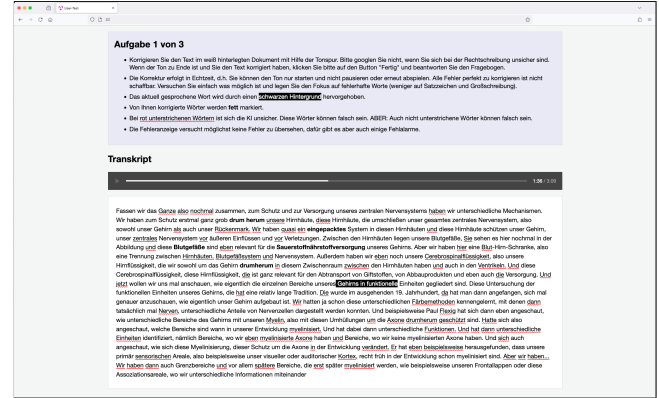


Figure 1: Correction interface of the user study.

and deletions, enabling classification of user corrections. User behaviour, such as typing speed and input events, was recorded with timestamps.

3.3.5 Questionnaire. After each task, participants completed a questionnaire with 7-point Likert scales. They rated transcription quality and answered four pragmatic quality questions from the short version of the User Experience Questionnaire (UEQ-S) [50], focusing on the efficiency of the interface rather than hedonic aspects. For interfaces with error highlighting, additional scales assessed the frequency, correctness, and usefulness of highlighted errors. An open comment field allowed detailed feedback. After the final task, participants reviewed all questionnaires, adjusted their ratings if needed, and answered which interface option they preferred.

3.3.6 Procedure. Participants accessed the prototype via a website containing instructions, interfaces, and questionnaires. They were informed about the study's goals, provided consent, and confirmed that they had no hearing, visual, reading or writing impairments. A technical check ensured audio functionality. Each participant completed three correction tasks, with a fixed transcript order and randomised interface order. Before each task, users received a brief introduction, including explanations regarding the error highlighting functionality (e.g., that the threshold tries to balance or to cover most errors). Participants were informed that highlighted words were not always incorrect and unhighlighted words were not always correct. Editing began with the start of the audio playback, followed by a questionnaire after each task. The study concluded with a final questionnaire and participant debriefing.

4 Results

4.1 General Analysis

4.1.1 Dataset Distribution. An Anderson-Darling test was conducted to check if the distribution of word-level confidence scores followed a normal distribution. The test result showed a statistic of 74901.069, exceeding the 0.787 critical value at the 5% significance level. A skewness test (statistic: -458.746, $p < .001$) showed a significant negative skew, with frequent scores clustered towards the high confidence scores. We conclude that the dataset is not normally distributed and has significant long tail and right skew

²<https://github.com/jitsi/jiwer>

³<https://github.com/ueberdosis/tiptap>

Table 1: Transcription accuracy, confidence correlation, and classification metrics across ASR models.

Model	WER↓	Confidence	r_s ↓	r_{pb} ↑	Threshold	SD↓	Precision↑	Recall↑	F1↑	AUC↑
Amazon	0.08	0.98	-0.64	0.413	0.92	0.09	0.52	0.56	0.52	0.87
AssemblyAI	0.06	0.92	-0.42	0.274	0.55	0.16	0.46	0.36	0.33	0.78
Deepgram	0.12	0.94	-0.84	0.539	0.81	0.10	0.55	0.59	0.55	0.86
Google	0.24	0.94	-0.57	0.314	0.94	0.05	0.41	0.50	0.44	0.68
IBM	0.15	0.87	-0.88	0.515	0.77	0.05	0.52	0.60	0.54	0.84
Rev AI	0.08	0.95	-0.73	0.490	0.87	0.09	0.49	0.64	0.54	0.87
Speechmatics	0.05	0.99	-0.63	0.427	0.87	0.12	0.49	0.50	0.47	0.80
SpeechText.AI	0.13	0.98	-0.78	0.396	0.93	0.06	0.54	0.41	0.46	0.69
Whisper	0.08	0.94	-0.61	0.505	0.70	0.22	0.48	0.54	0.47	0.83

characteristics. This is as expected, as most words are correctly transcribed, and E2E models exhibit overconfidence. Consequently, non-parametric tests were chosen for the general analysis.

4.1.2 Transcript WER and Confidence Score. Table 1 shows the mean WER and confidence score of all transcripts by ASR model. Spearman’s rank correlation was computed to assess the relationship between transcript-level WER and confidence score. For all models, WER was significantly related to confidence score (see Table 1 column r_s , all $ps < 0.001$). The correlation was moderate to strong and varies between the models independently of their mean WER. It can be concluded that the mean confidence score is higher for transcripts with a lower WER.

4.1.3 Word Correctness and Confidence Score. A point-biserial correlation analysis was conducted for each ASR model to assess the relationship between confidence score and word correctness. For all models, confidence scores were significantly related to word correctness (see Table 1 column r_{pb} , all $ps < 0.001$). The correlation was weak to moderate. The results suggest that words with higher confidence scores are more likely to be correct.

4.1.4 Classification Performance. We used the F1-score to evaluate the classification performance of confidence scores in detecting transcription errors. This metric ignores true negatives, which are less relevant in this context, as the classification task focuses on error detection within a dataset that mostly contains correct samples. For each transcript, the threshold that achieved the highest F1-score was used. The optimal thresholds varied between models, and some models also showed considerable variability (see Table 1 columns Threshold, SD). The classification performance of precision, recall, F1-score, and area under the curve (AUC) are shown in Table 1. The models achieved precision scores ranging from 0.41 to 0.55, indicating a 50% chance that detected errors were actual errors. Recall scores ranged from 0.36 to 0.64, suggesting that approximately 50% of transcription errors were missed. F1-scores ranged from 0.33 to 0.55, reflecting poor to moderate classification performance, indicating that the classifier has poor to moderate performance, balancing precision and recall, but not excelling at either. The Receiver Operating Characteristic (ROC) AUC values, ranging from 0.68 to 0.87, indicate a moderate fit of the classifier.

4.2 User Study Results

4.2.1 Word Error Rate Reduction. Participants’ corrections reduced the WER on average by 0.47 ($SD = 0.16$) without error highlighting, 0.48 ($SD = 0.14$) with balanced-threshold highlighting, and 0.45 ($SD = 0.17$) with high-threshold highlighting. A repeated-measures ANOVA was performed to evaluate the effect of the interface on the relative WER reduction. Mauchly’s test indicated that the assumption of sphericity had been met, $\chi^2(2) = 1.61, p = .447$. The results show that the WER reduction was not significantly affected by the type of interface, $F(2, 70) = 0.27, p = .767, \eta^2 = .005$.

4.2.2 Modification Ratio of Highlighted Errors. Chi-squared tests of independence were performed to assess the relationship between highlighted errors and user modifications. For the balanced-threshold interface, highlighted errors were significantly more likely to be modified, $\chi^2(1) = 30.32, p < .001$, with 2.64 times higher odds (CI 95% [1.87, 3.73]). For the high-threshold interface, no significant association was found, $\chi^2(1) = 1.18, p = 0.278$, with 1.31 times higher odds (CI 95% [0.84, 2.03]).

4.2.3 Transcription Quality Rating. Participants rated transcription quality on a seven-point Likert scale (1 = very bad, 7 = very good). Mean ratings were 5.9 ($SD = 1.3$) without highlighting, 5.8 ($SD = 1.0$) with balanced-threshold highlighting, and 5.4 ($SD = 1.3$) with high-threshold highlighting. A repeated-measures ANOVA was performed to evaluate the effect of the interface on the perceived transcription quality. Mauchly’s test indicated that the assumption of sphericity had been met, $\chi^2(2) = 4.0, p = .136$. The results show that the perceived transcription quality was not significantly affected by the type of interface, $F(2, 70) = 1.71, p = .188, \eta^2 = .028$.

4.2.4 Interface Efficiency Rating. Participants rated pragmatic quality using four seven-point Likert scales from the UEQ-S. Mean scores were 1.47 ($SD = 1.16$) without highlighting, 1.51 ($SD = 1.17$) with balanced-threshold highlighting, and 1.41 ($SD = 1.08$) with high-threshold highlighting. A repeated-measures ANOVA was performed to evaluate the effect of the interface on the UEQ-S pragmatic quality. Mauchly’s test indicated that the assumption of sphericity had been violated, $\chi^2(2) = 9.97, p = .007$, therefore Greenhouse-Geisser corrected tests are reported ($\epsilon = .80$). The results show that the UEQ pragmatic quality was not significantly affected by the type of interface, $F(2, 70) = 0.21, p = .765, \eta^2 = .001$.

4.2.5 Usefulness of Error Highlighting. Participants rated error highlighting on three seven-point Likert scales: frequency (1 = far too few, 7 = far too many), correctness (1 = very wrong, 7 = very correct), and usefulness (1 = very irritating, 7 = very helpful). Paired t-tests showed significantly better ratings for the balanced threshold compared to the high threshold condition in frequency ($M = 4.2$ vs. $M = 5.3$, $t(35) = -4.84$, $p < .001$), correctness ($M = 3.6$ vs. $M = 2.9$, $t(35) = 2.58$, $p = .014$), and usefulness ($M = 4.1$ vs. $M = 3.3$, $t(35) = 2.60$, $p = .014$).

4.2.6 Preferred Interface Type. After completing all tasks, participants rated their preferred interface. Two participants had no preference. 16 participants preferred no error highlighting, 18 balanced-threshold highlighting, and 4 high-threshold highlighting. In the optional comments section, 16 participants provided feedback on the error highlighting feature. Three expressed initial optimism about confidence-based highlighting before using it, 10 found it irritating, and 7 reported ignoring it after some time.

5 Discussion

We evaluated E2E ASR models to assess the reliability of confidence scores in detecting transcription errors. The results showed a strong correlation between WER and confidence at the transcript level, with higher confidence scores associated with fewer errors. However, confidence and word correctness showed only a weak to moderate correlation, highlighting the challenge of accurately classifying errors at the word level.

Setting a confidence threshold to classify word correctness is challenging due to the heavily skewed dataset. Most words have high confidence scores, and the confidence ranges for correct and incorrect words largely overlap. Thus, low thresholds miss most errors, while high thresholds produce many false positives. The inability to find a balanced threshold has already been reported for HMM-based ASR, and our results indicate that E2E models show similar limitations (cf. [16]).

We calculated the most effective threshold for each transcript based on the performance of the F1-score. The average threshold varied between and within the ASR models. This variance will likely increase in real-world scenarios, as our evaluation was limited to a single dataset. In practice, error classifiers would likely use suboptimal thresholds, as the optimal threshold is unknown during transcript correction. For the best thresholds, precision, recall, and F1-scores hovered around 0.5, indicating moderate success in identifying transcription errors. Precision shows that only half of the detections were actually errors, while recall shows that half of the errors were missed, highlighting the challenge of achieving high accuracy and coverage. ROC AUC values of 0.68 to 0.87 also indicate moderate classification performance, comparable to an HMM-based ASR evaluation study [57]. The low observed accuracy could lead users to distrust the system [67], and the presence of misses makes the classification unpredictable, a crucial aspect for the explainability of AI [61].

The user study showed that participants corrected about 50% of transcription errors, regardless of the error highlighting setting. Correction accuracy did not differ significantly between settings, suggesting that visual cues did not affect their ability to detect and correct errors. Highlighting effectively drew attention to words, as

long as highlighting was not too frequent (cf. [63]). Participants generally rated the transcription quality as high, and these ratings were not significantly affected by the interface configuration. Thus, error highlighting, regardless of its frequency or accuracy, did not influence overall confidence in transcription accuracy.

The UEQ-S results showed good pragmatic quality with no significant differences between settings, suggesting that the interface was perceived as efficient regardless of the presence or absence of error highlighting. In the balanced threshold condition, users rated the frequency, correctness and usefulness of highlighted errors positively. In the high threshold condition, highlighting was perceived as excessive, largely incorrect and irritating, indicating a preference for a more moderate approach. Most participants preferred either no highlighting or the balanced threshold condition. Similar to previous studies on keyword highlighting for DHH individuals [4, 55], participants found the concept of confidence-based error highlighting promising at first, but not very useful or even distracting after experiencing it.

6 Conclusion

Confidence scores have been proposed to automate error detection and assist users in correcting transcriptions. This study evaluated the reliability of these scores for error classification through a comprehensive analysis of current E2E ASR models and a user study examining their application in practice.

The results showed that confidence scores correlate with word correctness but are unreliable to guide manual correction. The overlap in confidence scores between correct and incorrect words, combined with the low proportion of incorrect words, makes it difficult to set a threshold and limits classification performance. Classifiers showed moderate success at best, failing to detect most errors or producing many false positives. The user study confirmed the general analysis, as confidence-based highlighting neither improved correction efficiency nor was perceived as helpful.

We conclude that current ASR confidence scores are insufficient for automated error detection and can be misleading, especially for those without a background in ML. This work aims to help HCI researchers understand the technical background, interpretation, and limitations of these scores in explainable AI.

7 Limitations & Future Work

The rapidly evolving field of ASR may lead to models and confidence measures that better reflect transcription errors. However, while E2E models have significantly improved ASR accuracy, our evaluation showed that confidence scores remain unreliable, suggesting a more fundamental problem. The results are limited to a single dataset, focusing on English and German under more favourable audio conditions, limiting generalisability to other languages. However, more extreme scenarios are likely to degrade classification performance [41]. Although the user study examined a real-time editing scenario, the classifier also appears ineffective for non-live correction tasks where users iteratively correct transcripts [39].

ASR is a valuable accessibility tool, but cannot signal uncertainty like human interpreters. Future research should focus not only on accuracy, but also on how ASR can communicate errors to increase the explainability of the results.

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References

- [1] Alexei Baevski, Yuhao Zhou, Abdelrahman Mohamed, and Michael Auli. 2020. wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. In *Advances in Neural Information Processing Systems*, Hugo Larochelle, Marc'Aurelio Ranzato, Raia Hadsell, Maria Florina Balcan, and Hsuan-Tien Lin (Eds.), Vol. 33. Curran Associates, Inc., virtual, 12449–12460. https://proceedings.neurips.cc/paper_files/paper/2020/file/92d1e1eb1cd6f9ba3227870bb6d7f07-Paper.pdf
- [2] Yassine Benayed, Dominique Fohr, Jean Paul Haton, and Gérard Chollet. 2003. Confidence measures for keyword spotting using support vector machines. In *2003 IEEE International Conference on Acoustics, Speech, and Signal Processing, 2003. Proceedings. (ICASSP '03)*, Vol. 1. IEEE, Hong Kong, China, I–I. doi:10.1109/ICASSP.2003.1198849
- [3] Larwan Berke. 2017. Displaying confidence from imperfect automatic speech recognition for captioning. *SIGACCESS Access. Comput.* 117 (feb 2017), 14–18. doi:10.1145/3051519.3051522
- [4] Larwan Berke, Christopher Caulfield, and Matt Huenerfauth. 2017. Deaf and Hard-of-Hearing Perspectives on Imperfect Automatic Speech Recognition for Captioning One-on-One Meetings. In *Proceedings of the 19th International ACM SIGACCESS Conference on Computers and Accessibility* (Baltimore, Maryland, USA) (ASSETS '17). Association for Computing Machinery, New York, NY, USA, 155–164. doi:10.1145/3132525.3132541
- [5] Moira Burke, Brian Amento, and Philip Isenhour. 2006. Error correction of voicemail transcripts in SCANMail. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Montréal, Québec, Canada) (CHI '06). Association for Computing Machinery, New York, NY, USA, 339–348. doi:10.1145/1124772.1124823
- [6] Janine Butler. 2019. Perspectives of deaf and hard of hearing viewers of captions. *American Annals of the Deaf* 163, 5 (2019), 534–553. <https://www.jstor.org/stable/26663593>
- [7] Ho Yin Chan and Philip C. Woodland. 2004. Improving broadcast news transcription by lightly supervised discriminative training. In *2004 IEEE International Conference on Acoustics, Speech, and Signal Processing*, Vol. 1. IEEE, Montreal, QC, Canada, I–737. doi:10.1109/ICASSP.2004.1326091
- [8] William Chan, Navdeep Jaitly, Quoc Le, and Oriol Vinyals. 2016. Listen, attend and spell: A neural network for large vocabulary conversational speech recognition. In *2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Shanghai, China, 4960–4964. doi:10.1109/ICASSP.2016.7472621
- [9] Lin Lawrence Chase. 1997. Word and acoustic confidence annotation for large vocabulary speech recognition.. In *Eurospeech*. ISCA, Rhodes, Greece, –.
- [10] Benjamin Chigier. 1992. Rejection and keyword spotting algorithms for a directory assistance city name recognition application. In *[Proceedings] ICASSP-92: 1992 IEEE International Conference on Acoustics, Speech, and Signal Processing*, Vol. 2. IEEE, San Francisco, CA, USA, 93–96 vol.2. doi:10.1109/ICASSP.1992.226112
- [11] Jan K Chorowski, Dzmitry Bahdanau, Dmitriy Serdyuk, Kyunghyun Cho, and Yoshua Bengio. 2015. Attention-Based Models for Speech Recognition. In *Advances in Neural Information Processing Systems*, Vol. 28. Curran Associates, Inc., Montreal, Canada, 9 pages. https://proceedings.neurips.cc/paper_files/paper/2015/file/1068c6e4c8051cfd4e9a8072e3189e2-Paper.pdf
- [12] Miguel Ángel Del-Agüa, Adrià Giménez, Albert Sanchis, Jorge Civera, and Alfons Juan. 2018. Speaker-Adapted Confidence Measures for ASR Using Deep Bidirectional Recurrent Neural Networks. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 26, 7 (2018), 1198–1206. doi:10.1109/TASLP.2018.2819900
- [13] Terrance DeVries and Graham W. Taylor. 2018. Learning Confidence for Out-of-Distribution Detection in Neural Networks. [arXiv:1802.04865 \[stat.ML\]](https://arxiv.org/abs/1802.04865)
- [14] Linhao Dong, Shuang Xu, and Bo Xu. 2018. Speech-Transformer: A No-Recurrence Sequence-to-Sequence Model for Speech Recognition. In *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Calgary, AB, Canada, 5884–5888. doi:10.1109/ICASSP.2018.8462506
- [15] Gunnar Evermann and Philip C. Woodland. 2000. Large vocabulary decoding and confidence estimation using word posterior probabilities. In *2000 IEEE International Conference on Acoustics, Speech, and Signal Processing. Proceedings (Cat. No.00CH37100)*, Vol. 3. IEEE, Istanbul, Turkey, 1655–1658 vol.3. doi:10.1109/ICASSP.2000.862067
- [16] Jinjuan Feng and Andrew Sears. 2004. Using confidence scores to improve hands-free speech based navigation in continuous dictation systems. *ACM Trans. Comput.-Hum. Interact.* 11, 4 (dec 2004), 329–356. doi:10.1145/1035575.1035576
- [17] Larry Gillick, Yoshiko Ito, and Jonathan Young. 1997. A probabilistic approach to confidence estimation and evaluation. In *1997 IEEE International Conference on Acoustics, Speech, and Signal Processing*, Vol. 2. IEEE, Munich, Germany, 879–882 vol.2. doi:10.1109/ICASSP.1997.596076
- [18] Anmol Gulati, James Qin, Chung-Cheng Chiu, Niki Parmar, Yu Zhang, Jiahui Yu, Wei Han, Shibo Wang, Zhengdong Zhang, Yonghui Wu, and Ruoming Pang. 2020. Conformer: Convolution-augmented Transformer for Speech Recognition. In *Interspeech 2020*. ISCA, virtual, 5036–5040. doi:10.21437/Interspeech.2020-3015
- [19] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 70)*. Doina Precup and Yee Whye Teh (Eds.), PMLR, Sydney, Australia, 1321–1330. <https://proceedings.mlr.press/v70/guo17a.html>
- [20] Timothy J. Hazen, Stephanie Seneff, and Joseph Polifroni. 2002. Recognition confidence scoring and its use in speech understanding systems. *Computer Speech & Language* 16, 1 (2002), 49–67. doi:10.1006/csla.2001.0183
- [21] Matthias Hein, Maksym Andriushchenko, and Julian Bitterwolf. 2019. Why relu networks yield high-confidence predictions far away from the training data and how to mitigate the problem. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. IEEE, Long Beach, CA, USA, 41–50.
- [22] Dan Hendrycks and Kevin Gimpel. 2018. A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks. [arXiv:1610.02136 \[cs.NE\]](https://arxiv.org/abs/1610.02136)
- [23] Jiaji Huang, Rewon Child, Vinay Rao, Hairong Liu, Sanjeev Satheesh, and Adam Coates. 2016. Active learning for speech recognition: The power of gradients. In *The 30th Conference on Neural Information Processing Systems (NeurIPS)*. Curran Associates, Inc., Barcelona, Spain, 1–5.
- [24] Yan Huang, Dong Yu, Yifan Gong, and Chaojun Liu. 2013. Semi-supervised GMM and DNN acoustic model training with multi-system combination and confidence re-calibration. In *Proc. Interspeech 2013*. ISCA, Lyon, France, 2360–2364. doi:10.21437/Interspeech.2013-551
- [25] Sushant Kafle and Matt Huenerfauth. 2018. A Corpus for Modeling Word Importance in Spoken Dialogue Transcripts. In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*. European Language Resources Association (ELRA), Miyazaki, Japan, 99–103. <https://aclanthology.org/L18-1016>
- [26] Sushant Kafle, Peter Yeung, and Matt Huenerfauth. 2019. Evaluating the Benefit of Highlighting Key Words in Captions for People who are Deaf or Hard of Hearing. In *Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility* (Pittsburgh, PA, USA) (ASSETS '19). Association for Computing Machinery, New York, NY, USA, 43–55. doi:10.1145/3308561.3353781
- [27] Kaustubh Kalgaonkar, Chaojun Liu, Yifan Gong, and Kaisheng Yao. 2015. Estimating confidence scores on ASR results using recurrent neural networks. In *2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, South Brisbane, Australia, 4999–5003. doi:10.1109/ICASSP.2015.7178922
- [28] Alexandros Kastanos, Anton Ragni, and Mark Gales. 2020. Confidence Estimation for Black Box Automatic Speech Recognition Systems Using Lattice Recurrent Neural Networks. In *ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Barcelona, Spain, 6329–6333. doi:10.1109/ICASSP40776.2020.9053264
- [29] Joseph Keshet, David Grangier, and Samy Bengio. 2009. Discriminative keyword spotting. *Speech Communication* 51, 4 (2009), 317–329. doi:10.1016/j.specom.2008.10.002
- [30] Suyoun Kim, Takaaki Hori, and Shinji Watanabe. 2017. Joint CTC-attention based end-to-end speech recognition using multi-task learning. In *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, New Orleans, LA, USA, 4835–4839. doi:10.1109/ICASSP.2017.7953075
- [31] Korbinian Kuhn, Verena Kersken, Benedikt Reuter, Niklas Egger, and Gottfried Zimmermann. 2024. Measuring the Accuracy of Automatic Speech Recognition Solutions. *ACM Trans. Access. Comput.* 16, 4, Article 25 (jan 2024), 23 pages. doi:10.1145/3636513
- [32] Ankur Kumar, Sachin Singh, Dhananjaya Gowda, Abhinav Garg, Shatrughan Singh, and Chanwoo Kim. 2020. Utterance Confidence Measure for End-to-End Speech Recognition with Applications to Distributed Speech Recognition Scenarios. In *Proc. Interspeech 2020*. ISCA, Shanghai, China, 4357–4361. doi:10.21437/Interspeech.2020-3216
- [33] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. 2017. Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*, I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett (Eds.), Vol. 30. Curran Associates, Inc., Long Beach, CA, USA, 6405–6416. https://proceedings.neurips.cc/paper_files/paper/2017/file/9ef2ed4b7fd2c810847ffa5fa85bce38-Paper.pdf
- [34] Vladimir Iosifovich Levenshtein. 1966. Binary codes capable of correcting deletions, insertions, and reversals. 707–710 pages.
- [35] Mingkun Li and Ishwar K Sethi. 2006. Confidence-based active learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 28, 8 (2006), 1251–1261. doi:10.1109/TPAMI.2006.156

- [36] Qiuqia Li, PM Ness, Anton Ragni, and Mark Gales. 2019. Bi-directional Lattice Recurrent Neural Networks for Confidence Estimation. In *ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Brighton, UK, 6755–6759. doi:10.1109/ICASSP.2019.8683488
- [37] Qiuqia Li, David Qiu, Yu Zhang, Bo Li, Yanzhang He, Philip C. Woodland, Lianliang Cao, and Trevor Strohman. 2021. Confidence Estimation for Attention-Based Sequence-to-Sequence Models for Speech Recognition. In *ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Toronto, ON, Canada, 6388–6392. doi:10.1109/ICASSP39728.2021.9414920
- [38] Saturnino Luz, Masood Masoodian, and Bill Rogers. 2008. Interactive visualisation techniques for dynamic speech transcription, correction and training. In *Proceedings of the 9th ACM SIGCHI New Zealand Chapter's International Conference on Human-Computer Interaction: Design Centered HCI* (Wellington, New Zealand) (CHINZ '08). Association for Computing Machinery, New York, NY, USA, 9–16. doi:10.1145/1496976.1496978
- [39] Saturnino Luz, Masood Masoodian, Bill Rogers, and Chris Deering. 2008. Interface design strategies for computer-assisted speech transcription. In *Proceedings of the 20th Australasian Conference on Computer-Human Interaction: Designing for Habitus and Habitat* (Cairns, Australia) (OZCHI '08). Association for Computing Machinery, New York, NY, USA, 203–210. doi:10.1145/1517744.1517812
- [40] Emma J. McDonnell, Ping Liu, Steven M. Goodman, Raja Kushalnagar, Jon E. Froehlich, and Leah Findlater. 2021. Social, Environmental, and Technical Factors at Play in the Current Use and Future Design of Small-Group Captioning. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW2, Article 434 (oct 2021), 25 pages. doi:10.1145/3479578
- [41] Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua Dillon, Balaji Lakshminarayanan, and Jasper Snoek. 2019. Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems*, H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett (Eds.), Vol. 32. Curran Associates, Inc., Vancouver, BC, Canada, 14003–14014. https://proceedings.neurips.cc/paper_files/paper/2019/file/8558cb408c1d76621371888657d2eb1d-Paper.pdf
- [42] Daniel S. Park, Yu Zhang, Ye Jia, Wei Han, Chung-Cheng Chiu, Bo Li, Yonghui Wu, and Quoc V. Le. 2020. Improved Noisy Student Training for Automatic Speech Recognition. In *Proc. Interspeech 2020*. ISCA, Shanghai, China, 2817–2821. doi:10.21437/Interspeech.2020-1470
- [43] Yifan Peng, Siddharth Dalmia, Ian Lane, and Shinji Watanabe. 2022. Branchformer: Parallel MLP-Attention Architectures to Capture Local and Global Context for Speech Recognition and Understanding. In *Proceedings of the 39th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 162)*. PMLR, Baltimore, MD, USA, 17627–17643. <https://proceedings.mlr.press/v162/peng22a.html>
- [44] Agnès Piquard-Kipffer, Odile Mella, Jérémy Miranda, Denis Jouvett, and Luiza Orosanu. 2015. Qualitative investigation of the display of speech recognition results for communication with deaf people. In *Proceedings of SLPAT 2015: 6th Workshop on Speech and Language Processing for Assistive Technologies*, Jan Alexandersson, Ercan Altınsoy, Heidi Christensen, Peter Ljunglöf, François Portet, and Frank Rudzicz (Eds.). Association for Computational Linguistics, Dresden, Germany, 36–41. doi:10.18653/v1/W15-5107
- [45] David Qiu, Yanzhang He, Qiuqia Li, Yu Zhang, Lianliang Cao, and Ian McGraw. 2021. Multi-Task Learning for End-to-End ASR Word and Utterance Confidence with Deletion Prediction. In *Proc. Interspeech 2021*. ISCA, Brno, Czechia, 4074–4078. doi:10.21437/Interspeech.2021-1207
- [46] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine Mcleavey, and Ilya Sutskever. 2023. Robust Speech Recognition via Large-Scale Weak Supervision. In *Proceedings of the 40th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 202)*, Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (Eds.). PMLR, Honolulu, Hawaii, USA, 28492–28518. <https://proceedings.mlr.press/v202/radford23a.html>
- [47] Antti Rasmus, Mathias Berglund, Mikko Honkala, Harri Valpola, and Tapani Raiko. 2015. Semi-supervised Learning With Ladder Networks. In *Advances in Neural Information Processing Systems*, Vol. 28. Curran Associates, Inc., Montreal, Canada, 9 pages. https://proceedings.neurips.cc/paper_files/paper/2015/file/378a063b8fdb1db941e34f4bde584c7d-Paper.pdf
- [48] Alberto Sanchis, Alfons Juan, and Enrique Vidal. 2003. Estimating Confidence Measures for Speech Recognition Verification Using a-Smoothed Naive Bayes Model. In *Pattern Recognition and Image Analysis*, Francisco José Perales, Aurélio J. C. Campilho, Nicolás Pérez de la Blanca, and Alberto Sanfeliu (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 910–918.
- [49] Thomas Schaaf and Thomas Kemp. 1997. Confidence measures for spontaneous speech recognition. In *1997 IEEE International Conference on Acoustics, Speech, and Signal Processing*, Vol. 2. IEEE, Munich, Germany, 875–878 vol.2. doi:10.1109/ICASSP.1997.596075
- [50] Martin Schrepp, Andreas Hinderks, and Jörg Thomaschewski. 2017. Design and Evaluation of a Short Version of the User Experience Questionnaire (UEQ-S). *International Journal of Interactive Multimedia and Artificial Intelligence* 4 (01 2017), 103. doi:10.9781/ijimai.2017.09.001
- [51] Matthew Stephen Seigel and Philip C. Woodland. 2011. Combining information sources for confidence estimation with CRF models. In *Proc. Interspeech 2011*. ISCA, Florence, Italy, 905–908. doi:10.21437/Interspeech.2011-352
- [52] Matthew Stephen Seigel and Philip C. Woodland. 2014. Detecting deletions in ASR output. In *2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Florence, Italy, 2302–2306. doi:10.1109/ICASSP.2014.6854010
- [53] Matthew Stephen Seigel, Philip C. Woodland, and Mark Gales. 2013. A confidence-based approach for improving keyword hypothesis scores. In *2013 IEEE International Conference on Acoustics, Speech and Signal Processing*. IEEE, Vancouver, BC, Canada, 8565–8569. doi:10.1109/ICASSP.2013.6639337
- [54] Matthew Seita, Sooyeon Lee, Sarah Andrew, Kristen Shinohara, and Matt Huenenrath. 2022. Remotely Co-Designing Features for Communication Applications Using Automatic Captioning with Deaf and Hearing Pairs. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 460, 13 pages. doi:10.1145/3491102.3501843
- [55] Brent N. Shiver and Rosalee J. Wolfe. 2015. Evaluating Alternatives for Better Deaf Accessibility to Selected Web-Based Multimedia. In *Proceedings of the 17th International ACM SIGACCESS Conference on Computers & Accessibility* (Lisbon, Portugal) (ASSETS '15). Association for Computing Machinery, New York, NY, USA, 231–238. doi:10.1145/2700648.2809857
- [56] Yuchun Shu, Bo Hu, Yifeng He, Hao Shi, Longbiao Wang, and Jianwu Dang. 2024. Error Correction by Paying Attention to Both Acoustic and Confidence References for Automatic Speech Recognition. In *Interspeech 2024*. ISCA, Kos, Greece, 3500–3504. doi:10.21437/Interspeech.2024-1605
- [57] Bernhard Suhm, Brad Myers, and Alex Waibel. 2001. Multimodal error correction for speech user interfaces. *ACM Trans. Comput.-Hum. Interact.* 8, 1 (mar 2001), 60–98. doi:10.1145/371127.371166
- [58] Zehai Tu, Ning Ma, and Jon Barker. 2022. Unsupervised Uncertainty Measures of Automatic Speech Recognition for Non-intrusive Speech Intelligibility Prediction. In *Proc. Interspeech 2022*. ISCA, Incheon, Korea, 3493–3497. doi:10.21437/Interspeech.2022-10408
- [59] Gokhan Tur, Dilek Hakkani-Tür, and Robert E. Schapire. 2005. Combining active and semi-supervised learning for spoken language understanding. *Speech Communication* 45, 2 (2005), 171–186. doi:10.1016/j.specom.2004.08.002
- [60] Luis F. Uebel and Phil C. Woodland. 2001. Speaker adaptation using lattice-based MLLR.
- [61] Jasper van der Waa, Tjeerd Schoonderwoerd, Jurriaan van Diggelen, and Mark Neerincx. 2020. Interpretable confidence measures for decision support systems. *International Journal of Human-Computer Studies* 144 (2020), 102493. doi:10.1016/j.ijhcs.2020.102493
- [62] Sunil Vemuri, Philip DeCamp, Walter Bender, and Chris Schmandt. 2004. Improving speech playback using time-compression and speech recognition. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vienna, Austria) (CHI '04). Association for Computing Machinery, New York, NY, USA, 295–302. doi:10.1145/985692.985730
- [63] Keith Vertanen and Per Ola Kristensson. 2008. On the benefits of confidence visualization in speech recognition. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Florence, Italy) (CHI '08). Association for Computing Machinery, New York, NY, USA, 1497–1500. doi:10.1145/1357054.1357288
- [64] Mitch Weintraub, Françoise Beaufays, Ze'ev Rivlin, Yochai Konig, and Andreas Stolcke. 1997. Neural-network based measures of confidence for word recognition. In *1997 IEEE International Conference on Acoustics, Speech, and Signal Processing*, Vol. 2. IEEE, Munich, Germany, 887–890 vol.2. doi:10.1109/ICASSP.1997.596078
- [65] Daniel Willett, Andreas Worm, Christoph Neukirchen, and Gerhard Rigoll. 1998. Confidence measures for HMM-based speech recognition. In *International Conference on Spoken Language Processing (ICSLP)*, Vol. 98. ISCA, Sydney, Australia, 3241–3244.
- [66] Alejandro Woodward, Clara Bonnin, Issey Masuda, David Varas, Elisenda Bou-Balust, and Juan Carlos Riveiro. 2020. Confidence Measures in Encoder-Decoder Models for Speech Recognition. In *Proc. Interspeech 2020*. ISCA, Shanghai, China, 611–615. doi:10.21437/Interspeech.2020-2215
- [67] Ming Yin, Jennifer Wortman Vaughan, and Hanna Wallach. 2019. Understanding the Effect of Accuracy on Trust in Machine Learning Models. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–12. doi:10.1145/3290605.3300509
- [68] Dong Yu, Balakrishnan Varadarajan, Li Deng, and Alex Acero. 2010. Active learning and semi-supervised learning for speech recognition: A unified framework using the global entropy reduction maximization criterion. *Computer Speech & Language* 24, 3 (2010), 433–444. doi:10.1016/j.csl.2009.03.004
- [69] Kai Yu, Mark Gales, Lan Wang, and Philip C. Woodland. 2010. Unsupervised training and directed manual transcription for LVCSR. *Speech Communication* 52, 7 (2010), 652–663. doi:10.1016/j.specom.2010.02.014